ComplainEase - Online Complaint Registration and Management System

INTERNET PROGRAMMING LAB

AIM

To develop an online complaint registration and management website using HTML, CSS and JavaScript in which users can register, post complaints with supporting documents and a location marked on a map, track the complaint status and give feedback; officers can view assigned complaints, update the status and upload proof of work; and the administrator can generate officer IDs and passwords, create or delete accounts, assign complaints, change the password and hide features from users.

ALGORITHM

Step 1: Start. Load accounts, officer requests, complaints and feature settings from localStorage (seed demo data on the first run).

Step 2: Show the login page with the roles User, Officer and Admin. A user registers with name, mobile, username and password. An officer registration only sends a request to the admin.

Step 3: Authenticate the ID, password and role and open the dashboard of that role.

Step 4: User - Post Complaint: select the category, enter the description, attach a document and click on the map to mark the location (latitude and longitude are calculated). Validate the fields, create the complaint with status Submitted and show the complaint number.

Step 5: User - View Status: list the user's complaints with category, location link, current status, assigned officer, work note and proof of work. After resolution the user can submit a rating and comment as feedback.

Step 6: Admin - Officer accounts: for each registration request generate a unique Officer ID and a random password and create the officer account. Accounts can be viewed and deleted.

Step 7: Admin - Assign: select an officer for each complaint; the status changes to Assigned. Admin can also change the password, edit the contact information and hide the Feedback or Contact feature from the users.

Step 8: Officer - View Complaints: list only the complaints assigned to the logged-in officer with their map location.

Step 9: Officer - Update: choose the complaint, set the status (In Progress / Resolved), write a work note, upload the proof-of-work photo (read as a data URL) and save.

Step 10: Save all changes to localStorage and refresh the tables. Logout returns to the login page. Stop.

PROGRAM

Note: because the programs run offline, the "Mark location" control is a clickable map panel that converts the clicked point into latitude and longitude and generates an "Open in Google Maps" link for it. In an online deployment this panel can be replaced by the Google Maps JavaScript API with the same latitude / longitude fields.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>ComplainEase - Online Complaint Management</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header><div class="logo">ComplainEase <small>Register &bull; Track &bull; Resolve</small></div>
  <div><span id="who"></span> <button id="logout" class="btn light hidden">Logout</button></div></header>
<main>
<section id="auth" class="card narrow">
  <h2 id="authTitle">Login</h2>
  <label>Login as<select id="role"><option value="user">User</option><option
      value="officer">Officer</option><option value="admin">Admin</option></select></label>
  <div id="regRows" class="hidden">
    <label>Full Name<input id="fullName"></label>
    <label>Mobile / Department<input id="extra" placeholder="Mobile (user) or Department (officer)"></label>
  </div>
  <label>Username / ID<input id="uname"></label>
  <label>Password<input id="pwd" type="password"></label>
  <button id="authBtn" class="btn">Login</button>
  <a href="#" id="toggleReg">New here? Register</a>
  <p class="hint">Demo: admin / admin123  |  ravi / user123  |  OFF101 / off123</p>
  <p id="authMsg" class="msg err"></p>
</section>

<!-- USER -->
<section id="userView" class="hidden">
  <div class="two">
    <div class="card"><h3>Post Complaint</h3>
      <label>Category<select id="cCat"><option>Road &amp; Potholes</option><option>Street
          Light</option><option>Water
          Supply</option><option>Garbage</option><option>Electricity</option></select></label>
      <label>Description<textarea id="cDesc" rows="3"></textarea></label>
      <label>Supporting Document / Photo<input type="file" id="cDoc"></label>
      <div id="mapWrap"><b>Mark location on map</b> <small>(click the map)</small>
        <div id="map"><div id="pin" class="hidden">&#128205;</div></div>
        <div id="coord">No location marked</div></div>
      <button id="postC" class="btn">Submit Complaint</button>
      <p id="userMsg" class="msg"></p></div>
    <div class="card"><h3>Feedback</h3>
      <div id="fbFeature">
        <label>Select resolved complaint<select id="fbSel"></select></label>
        <label>Rating<select id="fbRate"><option>5</option><option>4</option><option>3</option><option>2</option
            ><option>1</option></select></label>
        <label>Comments<input id="fbText"></label>
        <button id="fbBtn" class="btn alt">Send Feedback</button></div>
      <div id="contactFeature"><h3>Admin Contact Details</h3><div id="contactBox"></div></div></div>
  </div>
  <div class="card"><h3>My Complaints - Status Tracking</h3><div id="myC"></div></div>
</section>

<!-- OFFICER -->
<section id="officerView" class="hidden">
  <div class="card"><h3>Complaints Assigned to Me</h3><div id="offC"></div></div>
  <div class="card hidden" id="proofCard"><h3 id="proofTitle">Update Status &amp; Proof of Work</h3>
    <div class="grid"><label>Status<select id="stSel"><option>In
        Progress</option><option>Resolved</option></select></label>
    <label>Work note<input id="stNote"></label>
    <label>Proof of work (photo)<input type="file" id="proofFile" accept="image/*"></label></div>
    <button id="stBtn" class="btn">Update</button> <span id="offMsg" class="msg"></span></div>
</section>

<!-- ADMIN -->
<section id="adminView" class="hidden">
  <div class="two">
    <div class="card"><h3>Officer Registration Requests</h3><div id="reqTable"></div>
      <p id="credBox" class="msg"></p></div>
    <div class="card"><h3>Features &amp; Contact Information</h3>
      <label class="row"><input type="checkbox" id="ftFeedback"> Show Feedback to users</label>
      <label class="row"><input type="checkbox" id="ftContact"> Show Contact Details to users</label>
      <label>Contact details<input id="ctInfo"></label>
      <button id="saveSet" class="btn alt">Save Settings</button>
      <h3>Change My Password</h3>
      <label>New password<input id="newPw" type="password"></label><button id="chPw" class="btn
          alt">Change</button>
      <p id="admMsg" class="msg"></p></div>
  </div>
  <div class="card"><h3>All Accounts (create / delete)</h3><div id="accTable"></div></div>
  <div class="card"><h3>All Complaints - Assign to Officer</h3><div id="allC"></div></div>
```

```
</section>
</main>
<script src="script.js"></script>
</body>
</html>
```

style.css

```
:root{--c:#6a1b9a;--c2:#e8d5f2;--bg:#f6f1fa}
*{box-sizing:border-box}
body{margin:0;font-family:"Segoe UI",Arial,sans-serif;background:var(--bg);color:#2a2033}
header{display:flex;justify-content:space-between;align-items:center;background:linear-
    gradient(90deg,#4a148c,#8e44ad);color:#fff;padding:12px 26px}
.logo{font-size:24px;font-weight:700}.logo small{font-size:12px;font-weight:400;margin-left:8px;color:#e0c7f0}
main{max-width:1050px;margin:18px auto;padding:0 14px}
.card{background:#fff;border-radius:10px;padding:16px;margin-bottom:14px;box-shadow:0 1px 5px
    rgba(74,20,140,.15)}
.narrow{max-width:410px;margin:30px auto}
h2,h3{margin-top:0;color:var(--c)}
label{display:flex;flex-direction:column;font-size:13px;font-weight:600;margin-bottom:10px;gap:3px}
label.row{flex-direction:row;align-items:center;gap:8px}
input,select,textarea{padding:8px;border:1px solid #c9b3d8;border-radius:6px;font:inherit}
input[type=checkbox]{width:18px;height:18px}
.btn{background:var(--c);color:#fff;border:0;border-radius:6px;padding:8px 16px;cursor:pointer;font:inherit}
.btn.alt{background:#37474f}.btn.light{background:#fff;color:var(--
    c)}.btn.red{background:#c0392b}.btn.sm{padding:4px 10px;font-size:12px}
.two{display:grid;grid-template-columns:1fr 1fr;gap:14px}.grid{display:grid;grid-template-
    columns:repeat(3,1fr);gap:10px}
@media(max-width:760px){.two,.grid{grid-template-columns:1fr}}
.hint{font-size:12px;color:#665}
.msg{font-weight:700;color:#1e7d3a;min-height:18px}.msg.err{color:#c0392b}
.hidden{display:none!important}
table{width:100%;border-collapse:collapse;font-size:13px}
th,td{padding:7px;border-bottom:1px solid #e5d9ee;text-align:left;vertical-align:top}th{background:var(--c2)}
.badge{padding:2px 8px;border-radius:10px;color:#fff;font-size:11px;white-space:nowrap}
.Submitted{background:#d98a00}.Assigned{background:#2b6cb0}.InProgress{background:#8e44ad}.Resolved{background:#
    1e7d3a}
#map{position:relative;height:150px;margin:6px 0;border:2px solid var(--c);border-radius:8px;cursor:crosshair;
  background:linear-gradient(90deg,rgba(255,255,255,.55) 2px,transparent 2px) 0 0/40px 40px,
  linear-gradient(0deg,rgba(255,255,255,.55) 2px,transparent 2px) 0 0/40px 40px,
  linear-gradient(120deg,#cfe8c8 0 30%,#a9d3ee 30% 45%,#e8e2c4 45% 100%)}
#pin{position:absolute;font-size:26px;transform:translate(-50%,-100%);pointer-events:none}
#coord{font-size:12px;margin-bottom:8px}#coord a{color:var(--c)}
.proof{max-width:90px;border-radius:4px;display:block;margin-top:4px}
#contactFeature{margin-top:18px}
```

script.js

```
const $ = id => document.getElementById(id);
const STATUS = ['Submitted','Assigned','In Progress','Resolved'];
const cls = s => s.replace(' ','');
const seed = () => ({
  accounts:[{id:'admin',pw:'admin123',role:'admin',name:'Administrator'},
            {id:'ravi',pw:'user123',role:'user',name:'Ravi Kumar',extra:'9840011223'},
            {id:'OFF101',pw:'off123',role:'officer',name:'S. Meenakshi',extra:'Public Works'}],
  requests:[{name:'T. Karthik',dept:'Electricity Board',phone:'9840099887'}],
  complaints:[{id:1,user:'ravi',cat:'Street Light',desc:'Street light near bus stand not working for a week.',do
      c:'light.jpg',lat:13.0843,lng:80.2705,status:'Assigned',officer:'OFF101',note:'',proof:'',fb:null,date:'20
      26-10-18'}],
  settings:{feedback:true,contact:true,info:'Helpline: 1800-123-4567 | admin@complainease.in | Mon-Sat, 9 AM - 6
       PM'},
  nextC:2, nextOff:102});
let db = JSON.parse(localStorage.getItem('complainease') || 'null') || seed();
const save = () => localStorage.setItem('complainease', JSON.stringify(db));
let me = null, reg = false, pos = null, active = null;
const say = (el, t, bad) => { $(el).textContent = t; $(el).className = 'msg' + (bad ? ' err' : ''); };

/* ---------- authentication ---------- */
$('toggleReg').onclick = e => { e.preventDefault(); reg = !reg;
  $('authTitle').textContent = reg ? 'Registration' : 'Login'; $('authBtn').textContent = reg ? 'Register' :
      'Login';
  $('regRows').classList.toggle('hidden', !reg); $('toggleReg').textContent = reg ? 'Back to Login' : 'New here?
       Register'; };
$('authBtn').onclick = () => {
  const id = $('uname').value.trim(), pw = $('pwd').value, role = $('role').value;
  if(reg){
    if(role === 'admin') return say('authMsg','Admin accounts cannot self-register.',true);
    if(!$('fullName').value) return say('authMsg','Enter your full name.',true);
    if(role === 'officer'){
      db.requests.push({name:$('fullName').value, dept:$('extra').value, phone:''}); save();
      return say('authMsg','Request sent. The admin will generate your Officer ID and password.');
    }
    if(!id || !pw) return say('authMsg','Enter username and password.',true);
    if(db.accounts.some(a => a.id === id)) return say('authMsg','Username already exists.',true);
    db.accounts.push({id, pw, role:'user', name:$('fullName').value, extra:$('extra').value}); save();
    return say('authMsg','Registered successfully. Please login.');
  }
  const a = db.accounts.find(x => x.id === id && x.pw === pw && x.role === role);
  if(!a) return say('authMsg','Invalid ID, password or role.',true);
  me = a; render();
};
$('logout').onclick = () => { me = null; $('pwd').value = ''; render(); };

/* ---------- user: map, complaint, feedback ---------- */
$('map').onclick = e => {
  const r = $('map').getBoundingClientRect(), x = (e.clientX - r.left) / r.width, y = (e.clientY - r.top) /
      r.height;
  pos = {lat:+(13.12 - y*0.08).toFixed(5), lng:+(80.22 + x*0.10).toFixed(5)};      // map area around Chennai
  $('pin').style.left = (x*100) + '%'; $('pin').style.top = (y*100) + '%'; $('pin').classList.remove('hidden');
  $('coord').innerHTML = `Marked: ${pos.lat}, ${pos.lng} - <a target="_blank"
      href="https://www.google.com/maps?q=${pos.lat},${pos.lng}">Open in Google Maps</a>`;
};
$('postC').onclick = () => {
  if(!$('cDesc').value.trim()) return say('userMsg','Please describe the complaint.',true);
  if(!pos) return say('userMsg','Please mark the location on the map.',true);
  const f = $('cDoc').files[0];
  const c = {id:db.nextC++, user:me.id, cat:$('cCat').value, desc:$('cDesc').value, doc:f ? f.name : '-',
      lat:pos.lat, lng:pos.lng,
             status:'Submitted', officer:'', note:'', proof:'', fb:null, date:new
                 Date().toISOString().slice(0,10)};
  db.complaints.push(c); save(); $('cDesc').value = ''; $('pin').classList.add('hidden'); $('coord').textContent
       = 'No location marked'; pos = null;
  say('userMsg', `Complaint #${c.id} registered successfully.`); renderUser();
};
$('fbBtn').onclick = () => {
  const c = db.complaints.find(x => x.id == $('fbSel').value); if(!c) return;
  c.fb = {rate:$('fbRate').value, text:$('fbText').value}; save(); $('fbText').value = ''; say('userMsg','Thank
      you for your feedback.'); renderUser();
};
function renderUser(){
  $('fbFeature').classList.toggle('hidden', !db.settings.feedback);
  $('contactFeature').classList.toggle('hidden', !db.settings.contact);
  $('contactBox').textContent = db.settings.info;
  const mine = db.complaints.filter(c => c.user === me.id);
  $('fbSel').innerHTML = mine.filter(c => c.status === 'Resolved' && !c.fb).map(c => `<option
      value="${c.id}">#${c.id} - ${c.cat}</option>`).join('') || '<option value="">No resolved
      complaints</option>';
  $('myC').innerHTML =
      '<table><tr><th>ID</th><th>Category</th><th>Description</th><th>Location</th><th>Status</th><th>Officer /
      Work done</th></tr>' +
```

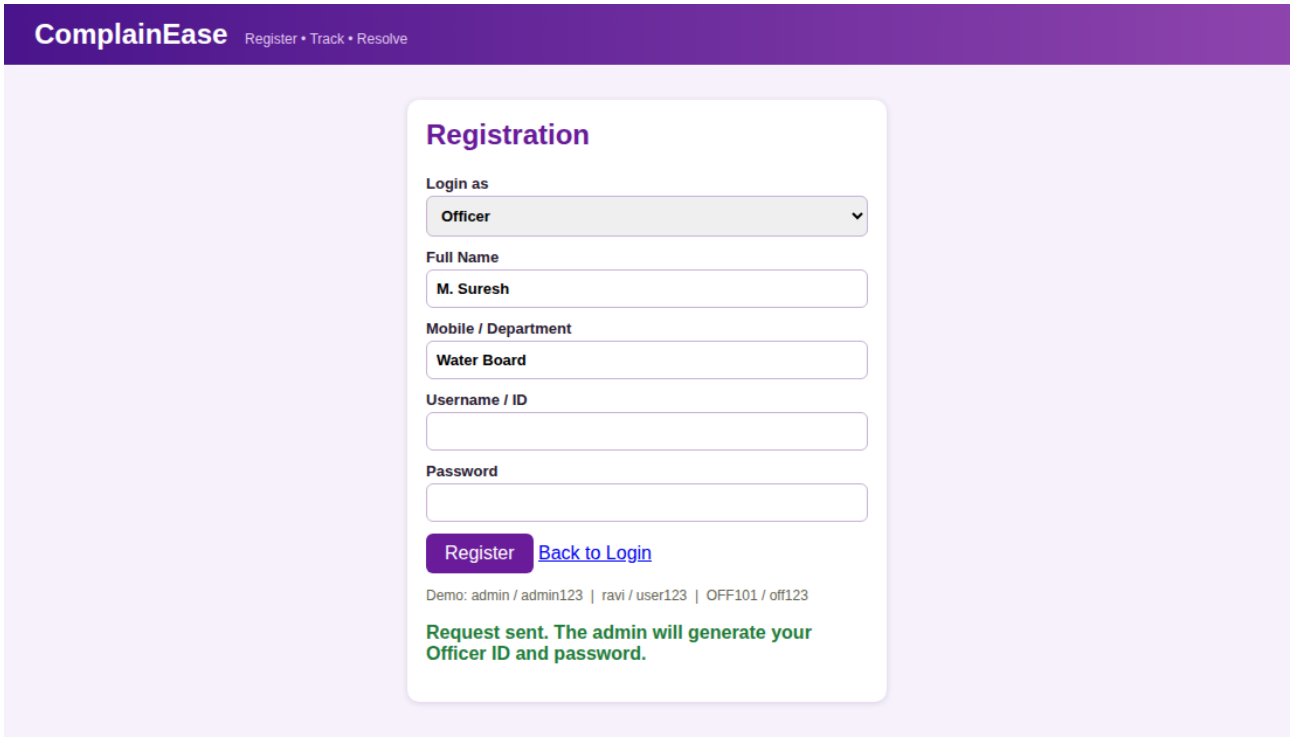
```
    mine.map(c => `<tr><td>#${c.id}</td><td>${c.cat}</td><td>${c.desc}<br><small>Doc: ${c.doc}</small></td>
    <td><a target="_blank" href="https://www.google.com/maps?q=${c.lat},${c.lng}">${c.lat}, ${c.lng}</a></td>
    <td><span class="badge ${cls(c.status)}">${c.status}</span></td>
    <td>${c.officer || 'Not assigned'}${c.note ? '<br>' + c.note : ''}${c.proof ? `<img class="proof"
        src="${c.proof}">` : ''}${c.fb ? `<br><small>Feedback: ${c.fb.rate}/5</small>` :
        ''}</td></tr>`).join('') + '</table>';
}

/* ---------- officer ---------- */
function renderOfficer(){
  const mine = db.complaints.filter(c => c.officer === me.id);
  $('offC').innerHTML = '<table><tr><th>ID</th><th>Category</th><th>Description</th><th>Location</th><th>Status<
      /th><th>Action</th></tr>' +
    mine.map(c => `<tr><td>#${c.id}</td><td>${c.cat}</td><td>${c.desc}</td><td><a target="_blank"
        href="https://www.google.com/maps?q=${c.lat},${c.lng}">${c.lat}, ${c.lng}</a></td>
    <td><span class="badge ${cls(c.status)}">${c.status}</span></td><td><button class="btn sm"
        onclick="pick(${c.id})">Update</button></td></tr>`).join('') + '</table>';
}
function pick(id){ active = id; $('proofCard').classList.remove('hidden'); $('proofTitle').textContent = `Update
     Status & Proof of Work - Complaint #${id}`; }
$('stBtn').onclick = () => {
  const c = db.complaints.find(x => x.id === active), f = $('proofFile').files[0];
  const done = proof => { c.status = $('stSel').value; c.note = $('stNote').value; if(proof) c.proof = proof;
      save(); say('offMsg', `Complaint #${c.id} updated to ${c.status}.`); renderOfficer(); };
  if(f){ const r = new FileReader(); r.onload = () => done(r.result); r.readAsDataURL(f); } else done('');
};

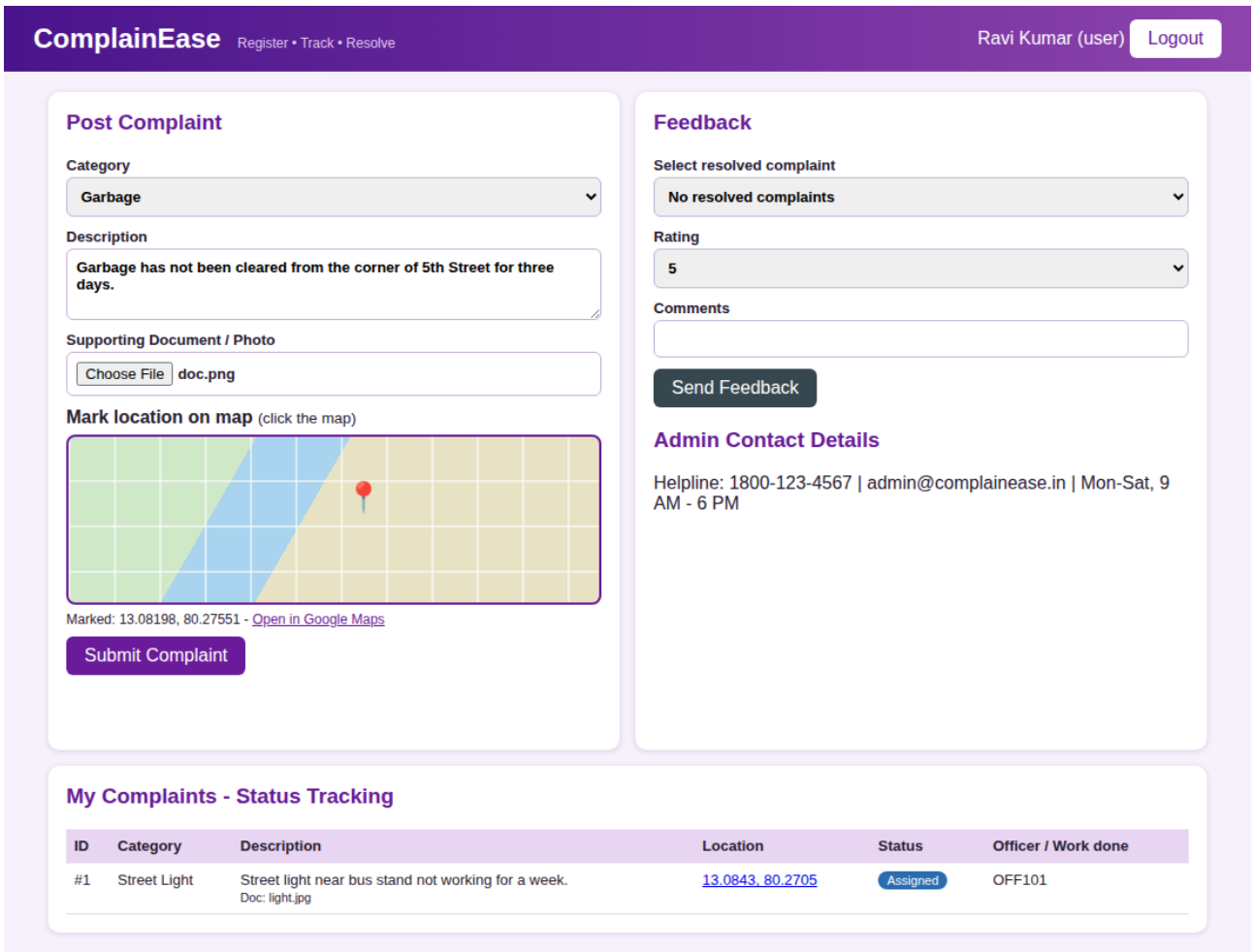
/* ---------- admin ---------- */
function genOfficer(i){
  const r = db.requests[i], id = 'OFF' + db.nextOff++, pw = 'pw' + Math.floor(1000 + Math.random()*9000);
  db.accounts.push({id, pw, role:'officer', name:r.name, extra:r.dept}); db.requests.splice(i, 1); save();
  $('credBox').textContent = `Generated for ${r.name}: ID = ${id}, Password = ${pw}`; renderAdmin();
}
function delAcc(id){ db.accounts = db.accounts.filter(a => a.id !== id); db.complaints.forEach(c => {
    if(c.officer === id){ c.officer = ''; c.status = 'Submitted'; } }); save(); renderAdmin(); }
function assign(id){
  const c = db.complaints.find(x => x.id === id); c.officer = $('as' + id).value; if(c.officer && c.status ===
      'Submitted') c.status = 'Assigned'; save(); renderAdmin();
}
$('saveSet').onclick = () => { db.settings = {feedback:$('ftFeedback').checked, contact:$('ftContact').checked,
    info:$('ctInfo').value}; save(); say('admMsg','Settings saved.'); };
$('chPw').onclick = () => { if(!$('newPw').value) return; me.pw = $('newPw').value; save(); $('newPw').value =
    ''; say('admMsg','Password changed.'); };
function renderAdmin(){
  $('ftFeedback').checked = db.settings.feedback; $('ftContact').checked = db.settings.contact;
      $('ctInfo').value = db.settings.info;
  $('reqTable').innerHTML = db.requests.length ?
      '<table><tr><th>Name</th><th>Department</th><th>Action</th></tr>' +
    db.requests.map((r, i) => `<tr><td>${r.name}</td><td>${r.dept}</td><td><button class="btn sm"
        onclick="genOfficer(${i})">Generate ID &amp;amp; Password</button></td></tr>`).join('') + '</table>' :
        '<p>No pending requests.</p>';
  $('accTable').innerHTML =
      '<table><tr><th>ID</th><th>Name</th><th>Role</th><th>Details</th><th>Action</th></tr>' +
    db.accounts.filter(a => a.role !== 'admin').map(a =>
        `<tr><td>${a.id}</td><td>${a.name}</td><td>${a.role}</td><td>${a.extra || ''}</td>
    <td><button class="btn sm red" onclick="delAcc('${a.id}')">Delete</button></td></tr>`).join('') +
        '</table>';
  const offs = db.accounts.filter(a => a.role === 'officer');
  $('allC').innerHTML =
      '<table><tr><th>ID</th><th>User</th><th>Category</th><th>Description</th><th>Status</th><th>Assign
      Officer</th></tr>' +
    db.complaints.map(c => `<tr><td>#${c.id}</td><td>${c.user}</td><td>${c.cat}</td><td>${c.desc}</td><td><span
        class="badge ${cls(c.status)}">${c.status}</span></td>
    <td><select id="as${c.id}"><option value="">--</option>${offs.map(o => `<option value="${o.id}" ${o.id ===
        c.officer ? 'selected' : ''}>${o.id} - ${o.name}</option>`).join('')}</select>
    <button class="btn sm" onclick="assign(${c.id})">Assign</button></td></tr>`).join('') + '</table>';
}

function render(){
  ['auth','userView','officerView','adminView'].forEach(s => $(s).classList.add('hidden'));
  $('logout').classList.toggle('hidden', !me); $('who').textContent = me ? `${me.name} (${me.role})` : '';
  if(!me) return $('auth').classList.remove('hidden');
  $(me.role + 'View').classList.remove('hidden');
  ({user:renderUser, officer:renderOfficer, admin:renderAdmin})[me.role]();
}
render();
```

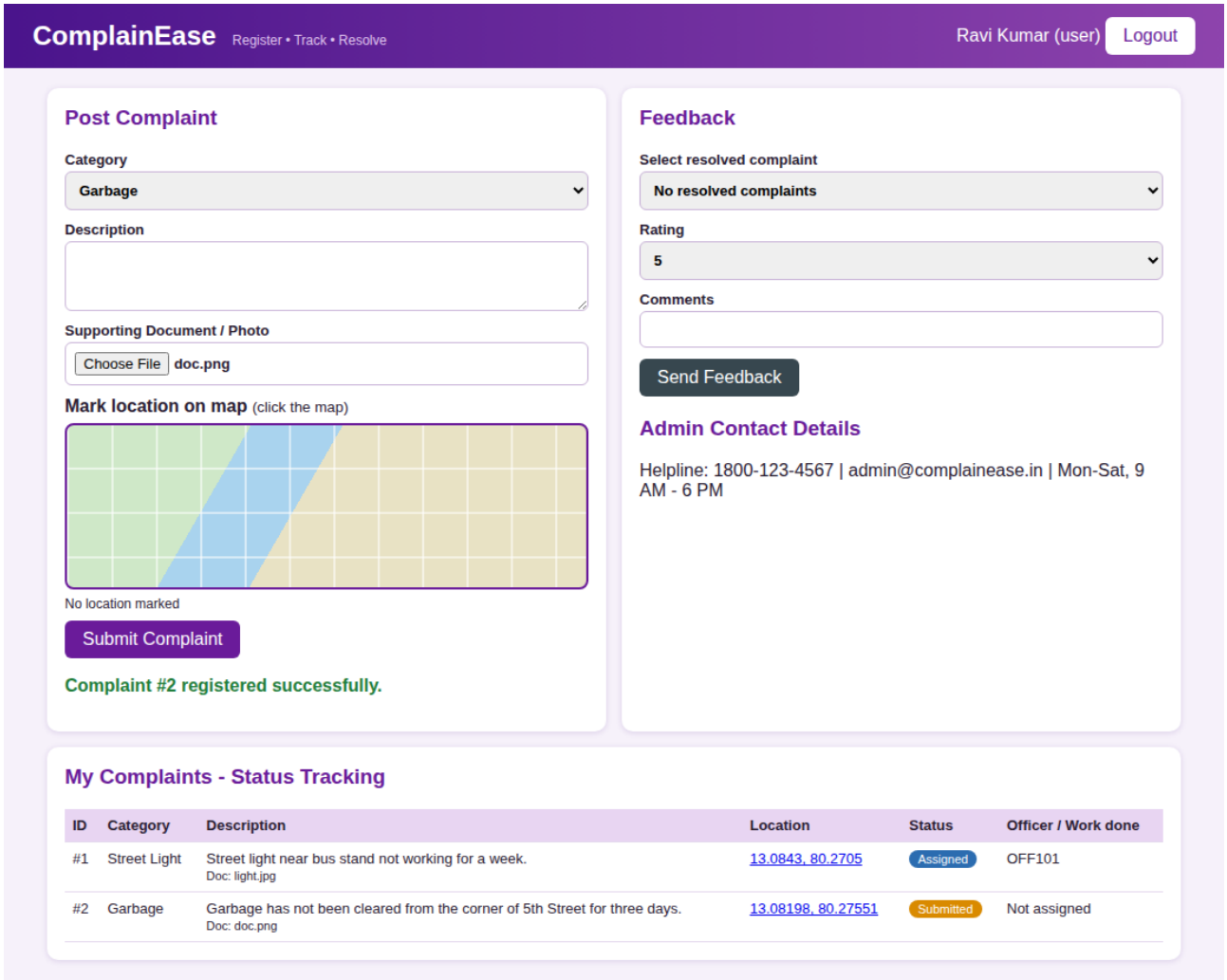
OUTPUT



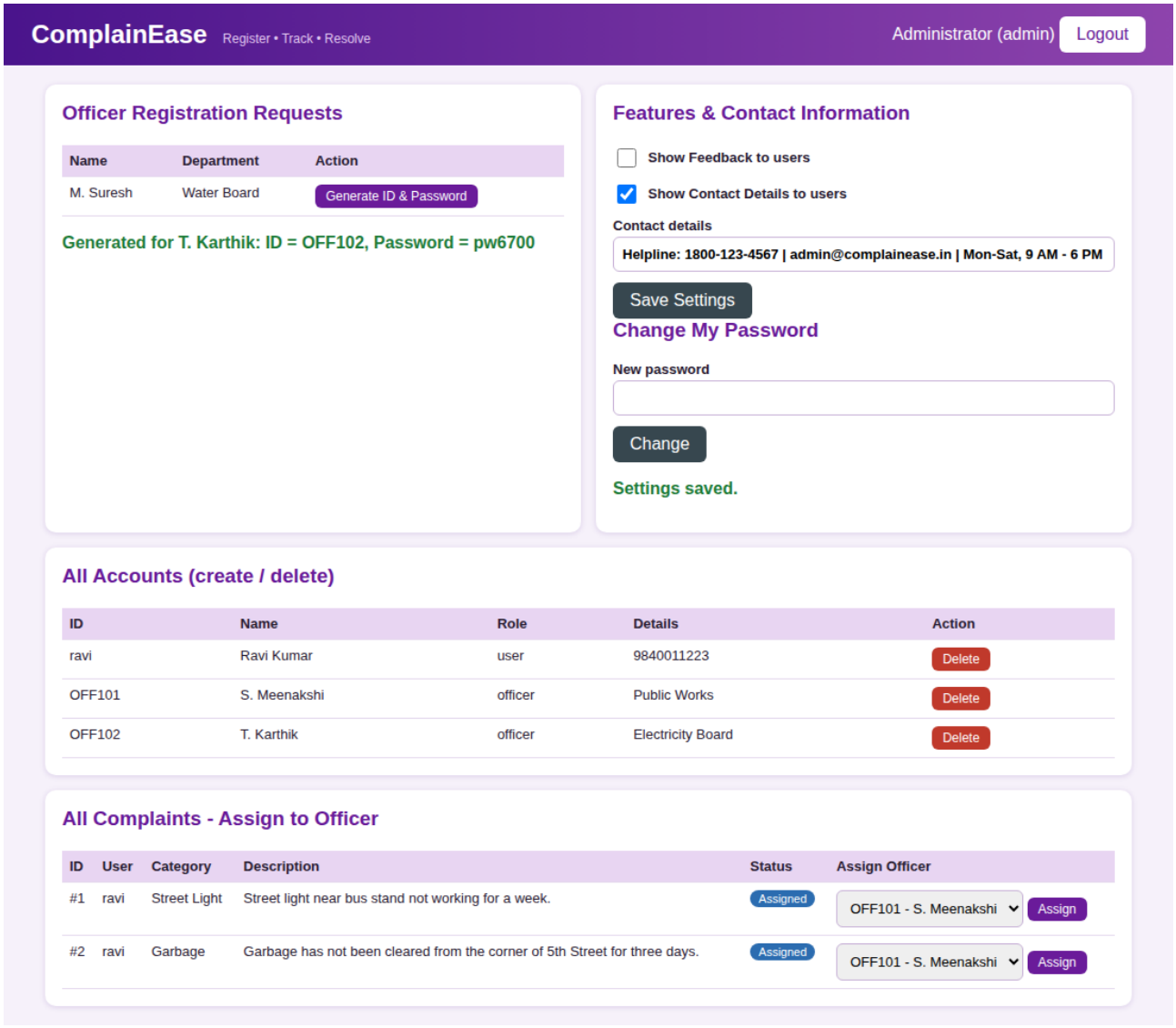
Registration page - an officer registration request is sent to the administrator



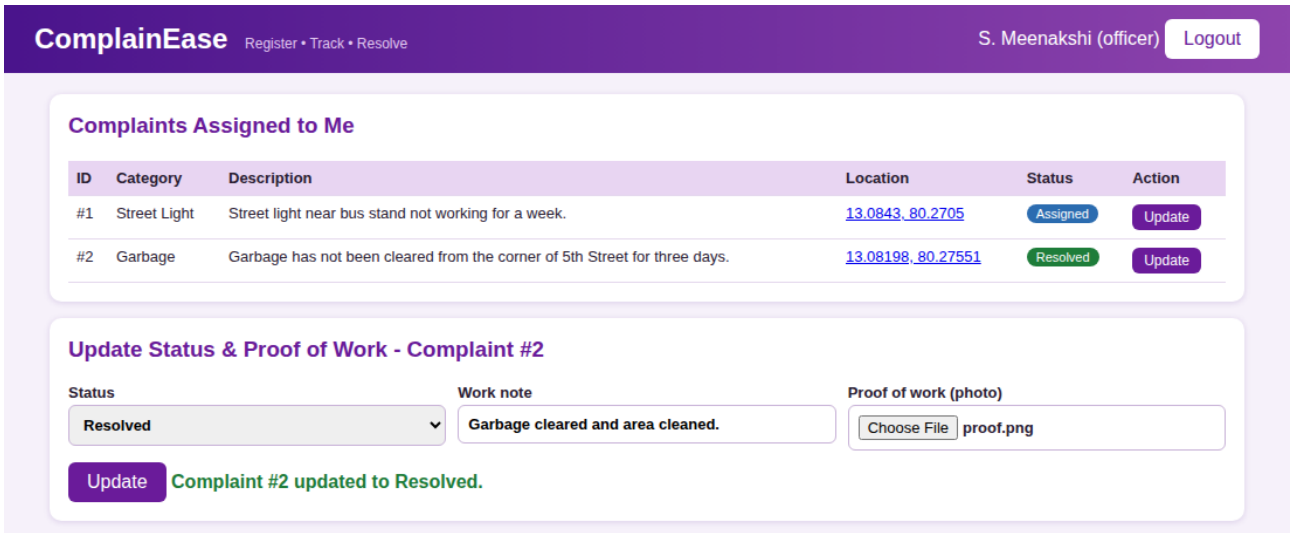
User posting a complaint: category, description, supporting document and the location marked on the map



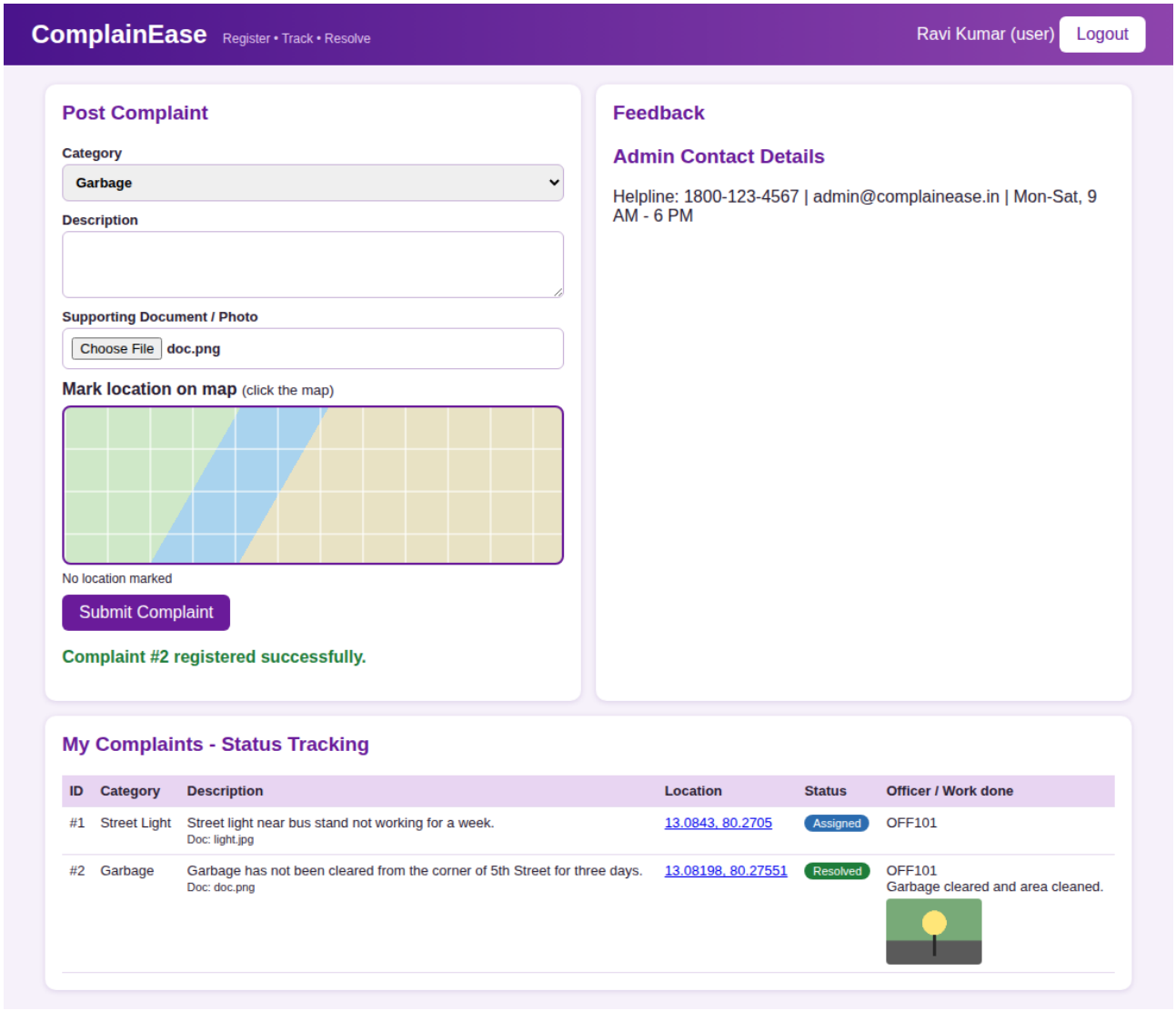
Complaint registered successfully and listed in the status tracking table



Admin dashboard: ID and password generated for the officer, complaint assigned, accounts managed and the Feedback feature hidden



Officer dashboard: viewing assigned complaints, updating the status and uploading the proof of work



User view after resolution: status Resolved with proof of work; Feedback section is hidden by the admin, contact details are shown

RESULT

The ComplainEase online complaint management website was developed using HTML, CSS and JavaScript. Users could post complaints with a map location and track their status, the admin could generate officer logins, manage accounts, assign complaints and hide features, and officers could update the status with proof of work. The output was verified successfully.